

1. Overview and Specifications

The Atlas-7 is an autonomous mobile picking robot for warehouses. It navigates with lidar and fiducial floor markers and is managed through the Fleet Console web application.

Key specifications: maximum payload 35 kg; maximum travel speed 2.0 meters per second; robot weight 118 kg; 48 V lithium iron phosphate (LiFePO₄) battery; runtime up to 10 hours per charge. Fast charging takes the battery from 0 to 80 percent in 45 minutes.

Operating environment: indoor use only, temperatures from 0 to 40 degrees Celsius, non-condensing humidity.

2. Safety

Each Atlas-7 has two red emergency stop buttons, one on each side of the chassis. Pressing either button cuts power to the drive motors immediately.

The safety lidar defines two protective zones. When a person or obstacle enters the slowdown zone, 1.5 meters from the robot, it reduces speed. If anything comes within the stop zone of 0.5 meters, the robot stops completely until the zone is clear.

Never ride on the robot or place items on top of the lidar housing. Before any maintenance, follow the lockout procedure: press an emergency stop, switch off the main power key, and attach your personal lock to the key switch.

3. Maintenance

Daily: inspect the drive wheels and casters for damage and debris.

Weekly: clean the lidar lens with a dry microfiber cloth. Do not use solvents.

Monthly: run the battery calibration routine from the Fleet Console.

Every 2,000 operating hours: replace both drive wheels. Wheel wear beyond this point increases localization errors.

Firmware updates are distributed through the Fleet Console and should be installed during a scheduled maintenance window.

4. Troubleshooting and Error Codes

E101 - Lidar obstruction: the lidar view is blocked. Clean the lens and remove any objects near the sensor.

E204 - Battery temperature high: move the robot to a cooler area and let it rest for 20 minutes before resuming operation.

E310 - Wheel motor stall: check the wheels for debris such as shrink wrap or straps, then clear the error in the Fleet Console.

E450 - Lost localization: the robot no longer knows its position. Manually drive it to the nearest fiducial marker and select Re-localize in the Fleet Console.

If an error persists, contact Northwind support at +1-412-555-0199.